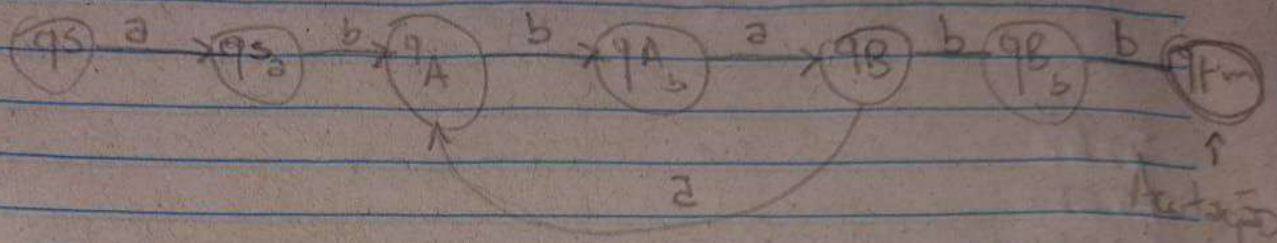
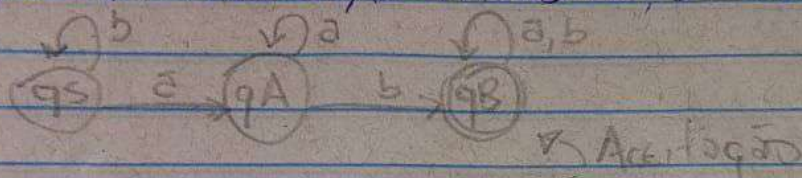


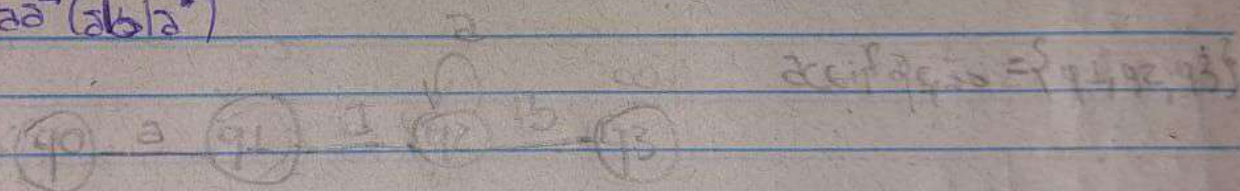
1) a. $D = \{S \rightarrow aA, A \rightarrow bA, B \rightarrow aA|b\}$



b) $P = \{S \rightarrow aA|bS, A \rightarrow aA|bB|b, B \rightarrow aB|bB|\epsilon\}$



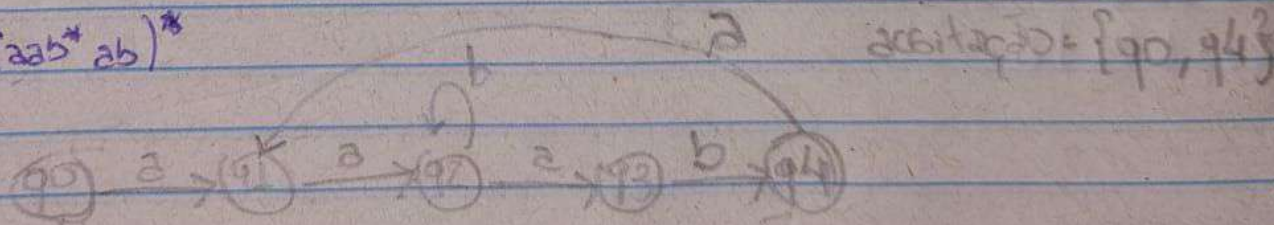
2) a) $a^*a^*(ab|a^*)^*$



$S \rightarrow aA$

$A \rightarrow aA|ab$

b) $(aab^*ab)^*$



$S \rightarrow aA|\epsilon$

$A \rightarrow aB$

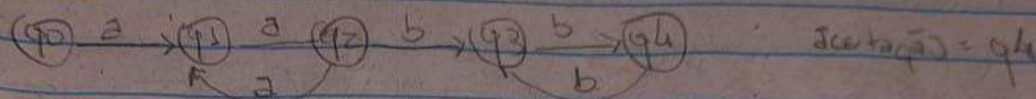
$B \rightarrow bB|aC$

$C \rightarrow b|baA$



D S T Q Q S S

c) $a^n b^m$, tal que n e m são pares (podem ser valores diferentes)



$S \rightarrow aA$

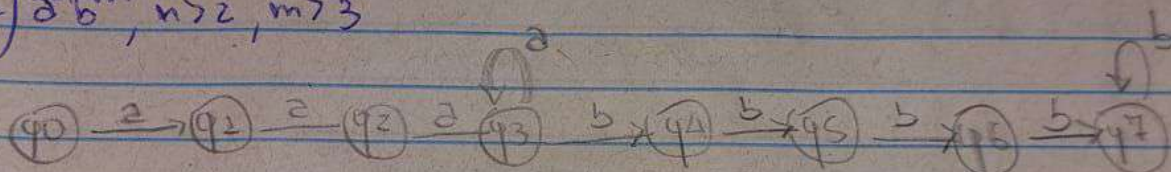
$A \rightarrow aA_1$

$A_1 \rightarrow aA \mid bB$

$B \rightarrow bB_1$

$B_1 \rightarrow bB \mid \epsilon$

d) $a^n b^m$, $n \geq 2$, $m \geq 3$



$S \rightarrow aA$

$A \rightarrow aA_1$

$A_1 \rightarrow aA_2$

$A_2 \rightarrow aA_2 \mid bB$

$B \rightarrow bB_1$

$B_1 \rightarrow bB_2$

$B_2 \rightarrow bB_3$

$B_3 \rightarrow bB_3 \mid \epsilon$